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Preparation and drug release property of tanshinone IIA loaded chitosan-montmorillonite microspheres

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ABSTRACT

In this study, a biocompatible chitosan/montmorillonite (CS/MMT) composite microsphere was developed as a carrier for loading and sustained-release of the hydrophobic drug of tanshinone IIA. Though the compatibility between hydrophobic drugs and hydrophilic matrix was fairly poor, tanshinone IIA was successfully loaded on the microsphere by the solvent exchange process during chitosan matrix dehydration. The microstructure of the resulting microspheres was characterized with several techniques, such as X-ray diffraction (XRD), thermogravimetric analysis (TGA), Fourier transform infrared spectroscopy (FT-IR), and scanning electron microscope (SEM). The results of drug loading and in vitro release study of the tanshinone IIA loaded CS/MMT composite microspheres showed that the incorporation of MMT into CS matrix would enhance the drug encapsulation and retard drug migration. The sample with mass ratio of CS: MMT (10:2) exhibited highest encapsulation efficiency (48.18% ± 2.54%) and slowest continuous cumulative release of drug in phosphate buffer solution (pH 7.4). It was found that the tanshinone IIA release kinetics fit the Higuchi model and the release mechanism was non-Fickian diffusion. Cell viability studies by CCK-8 assay showed that the microspheres showed no obvious cytotoxicity at the dosages below 80 μg/ml, and the MMT content had no significant effect on cell viability. This work provided a successful method of incorporating hydrophobic drugs into hydrophilic matrices, and has been successfully applied to the preparation of effective and biocompatible drug delivery for tanshinone IIA.
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1. Introduction

Tanshinone IIA (TA) is the major active ingredient isolated from the rhizome of the traditional Chinese medicine *Salvia miltiorrhiza* Bunge (Danshen) [1]. This hydrophobic compound (as shown in Fig. 1) is used to prevent and treat cardiovascular disorders, cerebrovascular diseases, neurodegenerative diseases, and cancer, and so on [2–8]. It is well known that many cardiovascular diseases need long-term therapy. However, the poor water-solubility (2.8 $ng \cdot ml^{-1}$) of TA [9] and short half-life (1–2 h) [10] often cause low oral bioavailability. To overcome these drawbacks, developing an efficient drug delivery system for TA should be a good solution, nanoparticles [11], polymeric micelles [12], solid dispersion [13], and micro-emulsions [14] have been used to improve the solubility of TA. Nevertheless, to the best of our knowledge, there are still few reports available on the sustained-release drug delivery system of TA.

The major difficulty in preparing a sustained-release drug delivery system is the poor compatibility between TA and carrier matrix, which is due to its hydrophobicity, leading to the low drug loading efficiency. Though the water-solubility of TA is very low, it has a much better solubility in ethyl alcohol, a well dehydrant for many polymer matrices, such as chitosan (CS), the ethanol molecule can enter into the polymer matrix via solvent exchange process of dehydration [15]. Enlightened from the above points, it is expected that the TA molecules dissolved in ethyl alcohol should be effectively loaded on the CS matrix during the dehydration process.

CS, a natural cationic polysaccharide derived from chitin, has been widely used for the matrix of drug delivery system due to its high biocompatibility, biodegradability, and non-toxicity [16,17]. Because of the abundant NH_2 and OH groups in CS backbone and its positively charged surface, CS can be easily prepared into different carrier materials, such as microspheres [18], nanoparticles [19], hydrogels [20], microgels [21], films [22], and other materials [23]. CS microspheres are used to provide sustained or controlled release of many drugs, and are able to improve the bioavailability of insoluble drugs, due to its well adhesive property of the mucosa and large surface to volume ratios. For example, Zhang et al. fabricated a novel phytosome-loaded CS microsphere system for curcumin delivery, which can promote the oral absorption, improve the bioavailability and prolong the retention time of curcumin in the body [24]. To achieve better comprehensive performances, pure CS drug carrier often requires further modifications with addition of other fillers. Among

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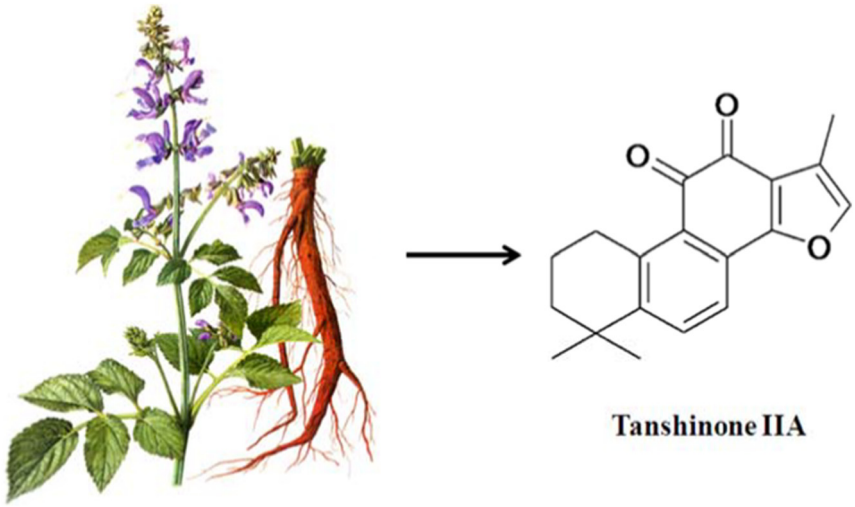
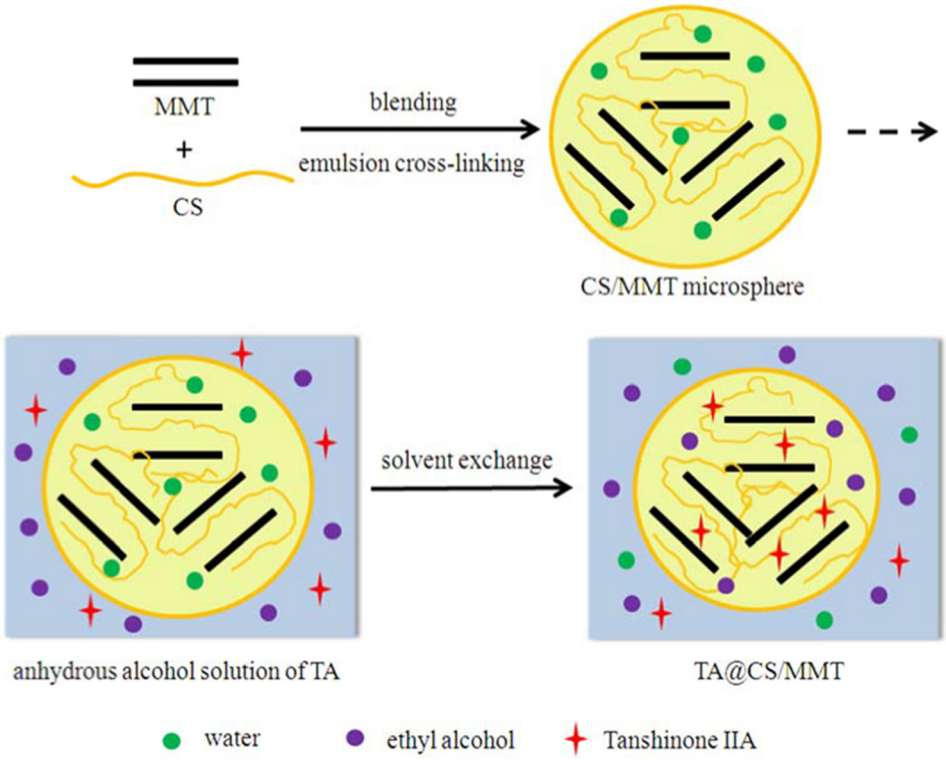


Fig. 1. The molecular structure of tanshinone IIA.

them, montmorillonite (MMT) clay particles have been widely used to improve the physicochemical and drug releasing properties in chitosan drug delivery system. MMT, a major constituent of bentonite, consists of layers of two tetrahedral silica sheets sandwiching one octahedral alumina sheet, is also widely applied in pharmaceuticals industry. The intercalation of some drugs onto MMT can overcome the problem of oral administration in clinical application [25]. Different scales of MMT material have been successfully used as drug carrier systems for sustained or controlled release [26–31]. Iannuccelli et al. use an organically modified MMT as drug delivery by intercalating tetracycline into MMT nanoplatelets, which can represent a potential advantageous approach allowing the antibiotic to distribute homogeneously on the gastric mucosa, diffuse through the gastric mucus layer and achieve the microorganism localization [32]. It is well known that CS has fairly good compatibility with MMT, the molecular chains of CS can effectively intercalate into Na^+-MMT layers, leading to an expansion of MMT interlayer spacing, which is advantageous for the drug entering into the interlayer spacing of MMT [33]. Moreover, the introduction of MMT into CS matrix can enhance the physical properties, i.e. mechanical strength, adsorption property, and stabilities of the composite [34,35]. CS/MMT microsphere is a common form of CS/MMT composite materials due to its simple and efficient preparation methods. And the common methods used for preparation CS/MMT microspheres include emulsion crosslinking [36], spray drying process [37], phase inversion [38], etc.

In the present work, the CS/MMT composite microspheres were fabricated through emulsion cross-linking technique, and the hydrophobic drug TA was successfully loaded on the microsphere by the solvent exchange process. The preparation process of the TA@CS/MMT is shown in Scheme 1: (i) dispersion of MMT into CS acetic acid solution and fabrication of CS/MMT composite microspheres through emulsion cross-



Scheme 1. Process of the preparation of the TA@CS/MMT.

linking technique, (ii) loading TA on the microsphere by the solvent exchange process. The microstructure of the resulting TA@CS/MMT microspheres was characterized with several techniques, such as X-ray diffraction (XRD), thermo-gravimetric analysis (TGA), Fourier transform infrared spectroscopy (FT-IR), and scanning electron microscope (SEM). The effects of varying proportion of MMT on the drug loading, encapsulation efficiency, and in vitro release kinetic were systematically studied. The human colon carcinoma cell line (Caco-2), an intestinal epithelial cell, is used for cytotoxicity test to verify the toxicity of microspheres. Compared with pure CS microspheres, CS/MMT composite microspheres are proven to have improved drug loading efficiency as well as slower drug releasing in vitro.

2. Experimental

2.1. Materials

Pharmaceutical grade, 95% deacetylated CS with a viscosity average molecular weight of 1.2×10^5, was supplied by Aoxing Biotechnology Co., Ltd. (Zhejiang, China). G-105 type of Na^+–MMT with cationic exchange capacity of 145 mmol/100 g was supplied by Nanocor Co., USA. Tanshinone IIA (98%) was provided from Hao Xuan biological Co., Ltd. (Xi'an, China). Span80 was purchased from TCI (Shanghai) Development Co., Ltd. (Shanghai, China). Glutaraldehyde was obtained from Energy Chemical, Sun Chemical Technology (Shanghai) Co., Ltd. The other used chemical reagents and solvents were supplied by Sinopharm Chemical Reagent Co., Ltd.

The human colon carcinoma cell line (Caco-2) was available from ATCC (American Type Culture Collection, USA). Gibco® Dulbecco's Modified Eagle Medium (DMEM, High Glucose, GlutaMAX™), Gibco® fetal bovine serum (FBS) were purchased from Thermo Fisher Scientific (USA). Nonessential amino acids were from Sigma-Aldrich Trade Co. (USA). Cell Counting Kit-8 (CCK-8) was purchased from Beyotime Biotechnology Co (Shanghai, China).

2.2. Preparation of TA@CS/MMT microspheres

CS/MMT microspheres were prepared using emulsion crosslinking method similar with previously work [39]. Briefly, 1 g of CS was dissolved in 50 ml 2 wt% acetic acid solution under magnetic stirring. Certain amounts (0, 0.1, 0.2, and 0.3 g, respectively) of MMT was dispersed in 25 ml 2 wt% acetic acid solution, and then added into the CS solution and stirred for another 4 h. The resultant mixture was injected through a syringe into 500 ml of oil phase containing span 80 (2%, v/v), and stirring was performed mechanically at 1000 rpm to form w/o emulsion at room temperature. The oil phase was a 7:5 (v/v) mixture of liquid-paraffin/cyclohexane. After 2 h of homogenization period, the obtained microspheres were cross-linked with 3 ml glutaraldehyde (50%, v/v) for 1 h. The crosslinked microspheres were collected by centrifugation (5000 rpm) and washed thrice with ethanol (50%, v/v). Afterwards, the excessive water of the microspheres was removed by filter paper.

A 0.5 g of the resultant microspheres was placed in an anhydrous ethanol solution containing 1 mg/ml of TA. After stirring for 24 h, the TA@CS/MMT microspheres were separated out and naturally dried. The residual TA in the solution was determined by HPLC at wavelength of 270 nm. The drug loading (DL) and encapsulation efficiency (EE) were calculated by the following equations. In addition, a group of dried CS/MMT microspheres (10:2) were used for the same adsorption experiments, to compare the efficiency of different drug loading methods.

$$\mathrm{DL}(\%) = \frac{M_0 - M_1}{M_2} \times 100\% \tag{1}$$

$$\mathrm{EE}(\%) = \frac{M_0 - M_1}{M_0} \times 100\% \tag{2}$$

where M_0 is the total amount of TA used for the formulation, M_1 is the amount of residual TA in the solution, M_2 is the total weight of the drug loaded microspheres. Three replicate tests were performed in the drug loading studies.

2.3. Characterization of TA@CS/MMT microspheres

The morphology of the CS/MMT microspheres was observed with a JEM-6360 scanning electron microscope (SEM). The average sizes and size distributions of the microspheres were determined based on the images of the particles using Nano Measurer 1.2 software. Fourier transform infrared spectroscopy (FTIR) analysis of the TA@CS/MMT microspheres was performed with a Nicolet 740 spectrometer, and the pellets were prepared by crushing samples with KBr. X-ray diffraction (XRD) analysis of the TA@CS/MMT microspheres was performed with an Empyrean X-ray diffraction system, the 2θ range of 5–60 ° was scanned with a speed of 2 °/min at room temperature. Thermal gravimetric (TG) analysis of the TA@CS/MMT microspheres was performed with a Mettler Toledo TGA/DSC 2 STARe system, under air atmosphere, ranging from 30 to 800 °C, at a scanning rate of 20 °C/min. The concentration of TA was determined using a Waters e2695-2489 liquid chromatographic system (HPLC) under the following conditions: SunFire C18 (4.6 mm × 250 mm, 5 μm) column, methanol and water (90:10 v/v) as mobile phase at a flow rate of 1.0 ml/min, UV detection at 270 nm.

2.4. In vitro release study

A dynamic dialysis method was employed. TA@CS/MMT equivalent to 2 mg TA were suspended in 5.0 ml of PBS (pH 7.4, 37 °C, containing 0.5% sodium dodecyl sulfate) and the dispersion was sealed in a dialysis bag (MWCO = 10,000) which was immersed in 195 ml of PBS. The permeated solution was stirred at a rate of 100 r/min. At the desired time intervals, 5 ml of the medium was extracted from the release medium to analyze the amount of TA released, and then added with 5 ml of fresh PBS. The preset intervals were 0.5, 1, 2, 4, 6, 8, 10, 12, 16 and 24 h.

The following formula is used to calculate the cumulative release amount of TA:

$$M_t = \mathrm{V}\mathrm{C}_n + \sum_{i=1}^{n-1} \mathrm{C}_i V_i \tag{3}$$

where M_t is the cumulative released amount, V is the total solution volume in the permeation pool, C_n is the solution concentration of TA corresponding to the *n*th extracting, C_i is the solution concentration corresponding to the *i*th extracting, V_i is the extracting volume of the permeated TA solution each time. Three replicate tests were performed in the in vitro drug release studies.

2.5. Evaluation of in-vitro release kinetics

Release mechanism of the TA was investigated by analyzing the release data with the following mathematical models: zero-order (Eq. (4)), first-order (Eq. (5)), Korsmeyer-Peppas model (Eq. (6)) and Higuchi model (Eq. (7)) [40]. Where Q is the percentage of drug released at time t; and k_0, k_1, k_H, and k_P are the rate constants of the Eqs. (4), (5), (6), and (7), respectively. Moreover, to confirm the drug release mechanism, the first 60% of drug release was fitted with Korsmeyer-Peppas model. The *n* value is used to characterize different release mechanisms. For the case of spherical matrices, the value of n below 0.43 for Fickian diffusion, 0.43–0.85 for non-Fickian transport and above 0.85 means a case-II transport [41].

$$Q = k_0 t \tag{4}$$

$$\ln(1 - Q) = k_1 t \tag{5}$$

$$Q = k_H t^1/2 \quad (6)$$

$$Q = k_P t^n \quad (7)$$

2.6. Cell culture and evaluation of cytotoxicity

Caco-2 cells were cultured in DMEM culture medium (pH 7.4) supplemented with glutamax, containing 20% (V/V) heat inactivated FBS and 1% (V/V) nonessential amino acids, maintaining at 37 °C, 5% CO_2 atmosphere. Fresh medium was replaced every 48 h, and cells were passaged when reaching 80% confluence.

The safety of CS/MMT and TA@CS/MMT microspheres were evaluated by the CCK-8 assay. Briefly, Caco-2 cells were seeded in 96-well plates at a density of 1×10^4 cells/0.1 ml and cultured for 24 h (37 °C, 5% CO_2). Then, the culture medium was replaced by 200 μl of basal medium with four different concentrations of microspheres dispersions (10, 50, 80, 100 μg/ml), with basal medium used as blank controls. The plates were then incubated (37 °C, 5% CO_2) for 24 h, and the medium was replaced by 90 μl of medium and 10 μl of CCK-8, and continue to incubate for another 4 h. Afterwards, the absorbance was assayed by ELISA plate reader (Thermo Scientific™ Multiskan™ FC) at wavelength of 450 nm. The cell viability was reflected as a percentage of absorbance relative to that of the untreated control. Five individual experiments were performed and analyzed in each group.

2.7. Statistical analysis

The results were represented as mean ± standard deviation (mean ± SD). Statistical analysis was performed with SPSS 17.0 software. One way analysis of variance (ANOVA) and LSD test were used for data analysis. $P < 0.05$ was considered as statistically significant.

3. Results and discussion

3.1. Characterization

3.1.1. Morphology of the microspheres

The morphology of CS and CS/MMT microspheres were investigated by SEM (Fig. 2). The CS and CS/MMT microspheres were in a shape of well-defined sphere with the mean diameter in about 11–14 μm. Due to the lower technological requirements of emulsifying crosslinking process, the introduction of MMT has little effect on the preparation conditions, morphology and size distribution of microspheres. As shown in Fig. 2a, the pure CS microsphere exhibits a smooth surface. Upon addition of MMT with different content, all the CS/MMT microspheres exhibit rough surfaces, indicating fairly good interfacial interaction between MMT particles and CS matrix. The dispersion size of MMT particles (≤5 nm) within CS matrix does not show obvious increase as the content of MMT particles increase, indicating ideal miscibility between MMT and CS. The stable and uniform dispersion of MMT phase in the chitosan matrix is an important requirement for the three-dimensional network structure of CS/MMT microspheres [42].

3.1.2. FT-IR spectra

The FT-IR spectra of TA@CS/MMT microspheres and other control samples is shown in Fig. 3. CS shows characteristic peaks at 1641 cm^{-1} (C═O stretching vibration (amide I)), 1609 cm^{-1} (N—H in-plane bending vibration (amide II)), 1095 cm^{-1} (C—O stretching vibration). For MMT, the intensity of characteristic peaks related to Si—O, Si—O—Si and Al—O at 1035, 920, 515, 486 cm^{-1} are weakened in the spectra of the CS/MMT microspheres, indicating the successful dispersion of MMT into CS matrix. As for the CS/MMT microspheres, the peak of N—H bending vibration of NH_2 group of CS at 1609 cm^{-1} shifts to 1570 cm^{-1}, which is attributed to the occurrence of hydrogen bonding interaction between the amino groups of CS and Si—OH groups of MMT, similar to other chitosan-clay composites [43]. The absorption

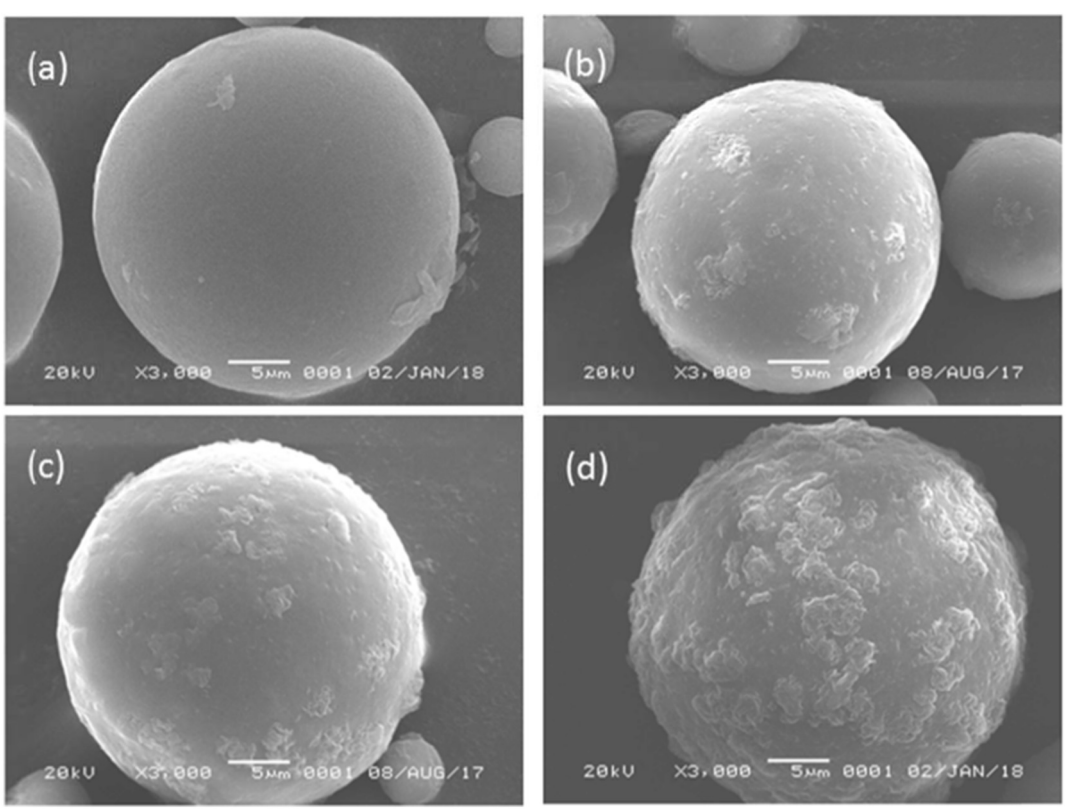


Fig. 2. SEM images of the microspheres: (a). CS; (b). CS/MMT(10:1); (c). CS/MMT(10:2); (d). CS/MMT(10:3).

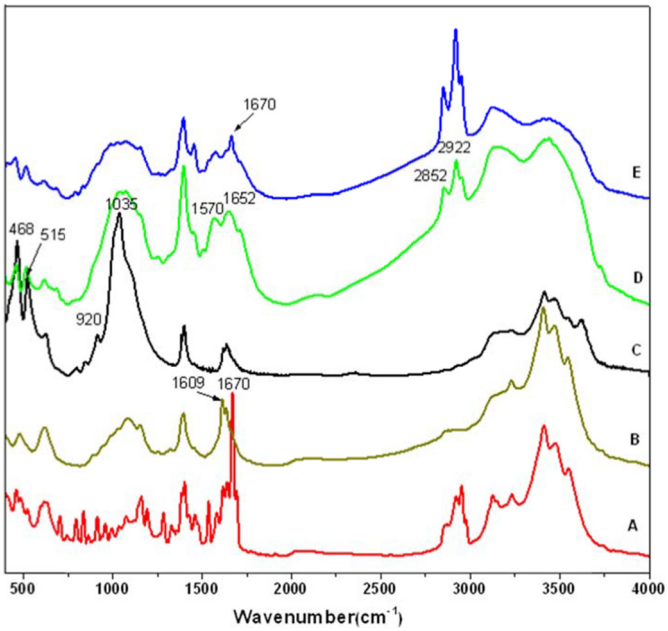


Fig. 3. FTIR spectrums of A. TA; B. CS; C. MMT; D. CS/MMT; and E. TA@CS/MMT.

peak at 1652 cm^{-1} for CS/MMT microspheres ascribes to the imine bonds —C═N of Schiff base, which indicates the cross-linked structure between CS and glutaraldehyde (as shown in Scheme 2) has been successfully established [44]. And the additional peaks at 2852, 2922 cm^{-1} (C—H stretching vibration of —CH and —CH_2 in glutaraldehyde) are also implied the cross-linked structure of microspheres. The main characteristic peak of TA is found at 1670 cm^{-1}, which is attributed to the C═O stretching vibration. And for TA@CS/MMT, the intensity of characteristic peak related to C═O is increased, indicating the successful loading of TA into CS/MMT microspheres.

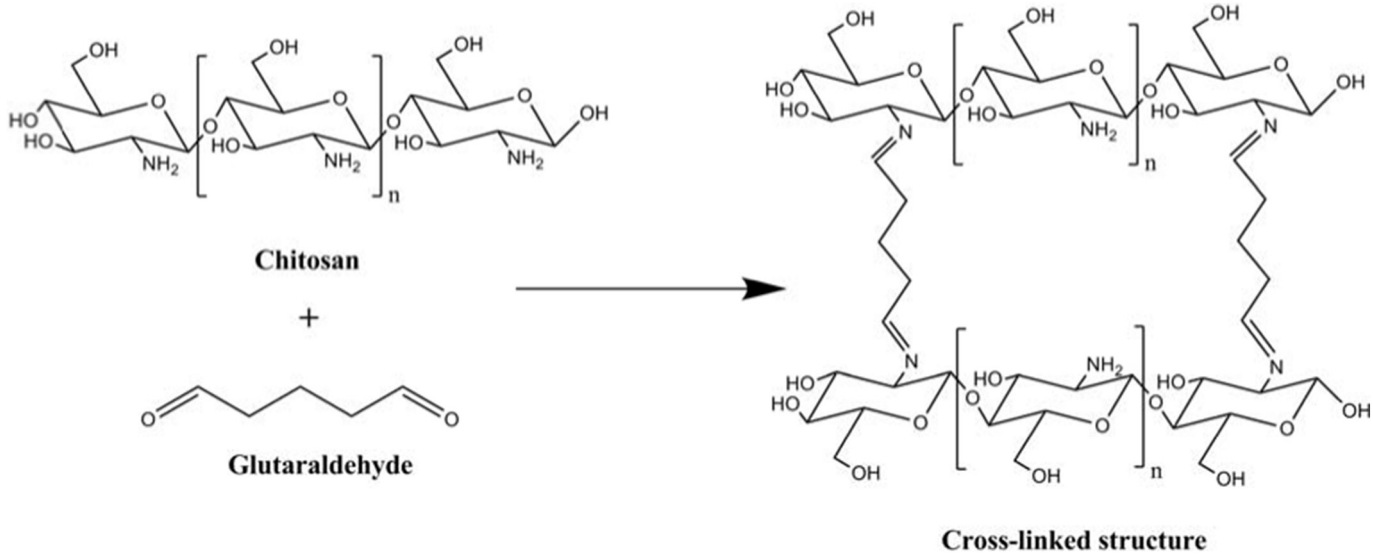
3.1.3. X-ray diffraction (XRD)

As seen, Fig. 4 represents the XRD patterns of CS, MMT, TA, CS/MMT microspheres, and TA@CS/MMT microspheres. The diffraction peaks of CS appear at 2θ of 10.77°, 20.12° are characteristic of the hydrated crystalline structure [45]. Pure MMT shows an intense characteristic reflection peak at 2θ of 7.08°, indicating the layer-to-layer spacing of 1.25 nm (d_{001} spacing). After the incorporation of MMT into CS matrix, the characteristic reflection peak of MMT is captured at 2θ of 5.95°, with corresponding d_{001} spacing of 1.48 nm. It indicates the layer-to-layer spacing of MMT is enlarged after intercalation of CS macromolecules into MMT interlayer spacing. And its intensity is largely weakened after forming the CS/MMT microspheres, suggesting that the MMT has dispersed at the molecular level in the polymer matrix [46]. After the drug was loaded, the characteristic reflection peak of TA@CS/MMT is found at 2θ of 5.49°, with corresponding d_{001} spacing of 1.61 nm. These results all suggest that TA molecules entered into the interlayer space of MMT. Pure TA shows an intense characteristic diffraction peak at 2θ of 7.20° due to the presence of TA crystals. The characteristic XRD peaks of TA crystals are observed in the TA@CS/MMT microspheres, confirming the success loading of TA on the CS/MMT microspheres.

3.1.4. Thermo gravimetric analysis (TGa)

The TGA-DTG curves of CS, MT, TA, CS/MMT, and TA@CS/MMT microspheres are shown in Fig. 5. MMT shows a high thermal stability containing only a small weight loss (weight loss ratio: about 15%) for the whole heating process (30 to 800 °C). It can be ascribed to the superficial humidity and the OH groups coordinated to the atoms in the octahedral layers. CS and CS/MMT microspheres have a weight loss of absorbed water around 100 °C, and the weight loss ratio is about 7%. Then the CS component begins to rapidly decompose in two sharp weight loss stages at 210–430 °C (weight loss ratio: about 62%), and 430–610 °C (weight loss ratio: about 31%), which are assigned to decomposition of the polymer via deacetylation-depolymerization and degradation of the final waste of the polymer, respectively. As for CS/MMT microspheres, it have multiple weight loss stages at 136–375 °C (weight loss ratio: about 35%), 375–558 °C (weight loss ratio: about 24%), 558–642 °C (weight loss ratio: about 16%). For comprehensive comparing, CS/MMT microspheres have much higher thermal stability than pure CS. The reason should be attributed to the MMT reinforcing effects and the formation of interactions between CS and MMT.

As shown in Fig. 5, TA underwent three stages of weight loss at 217–366 °C, 366–505 °C, 505–800 °C, and the mass loss of TA are calculated to be approximately 13, 18, and 69% respectively. After the addition of drugs, the TGA-DTG curve of TA@CS/MMT shows mild different from that of CS/MMT. The mass loss of the drug loaded microspheres at 360–700 °C is slower than CS/MMT microspheres, and the total weight loss is more than CS/MMT microspheres, which is consistent with the tendency of the TA TGA curve. These results confirm the presence of drugs in TA@CS/MMT microspheres.



Scheme 2. The mechanism of CS crosslinking.

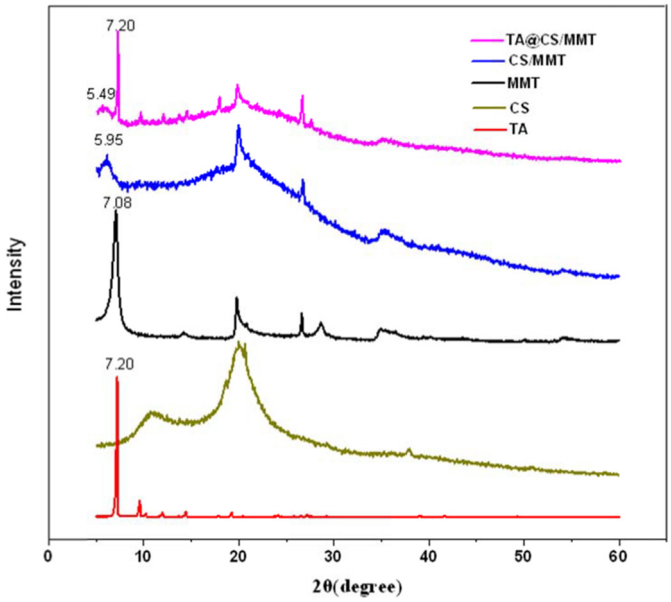


Fig. 4. XRD patterns of TA, CS, MMT, CS/MMT, and TA@CS/MMT.

3.2. Drug loading (DL) and encapsulation efficiency (EE) analysis

The efficiency of different drug loading methods, the influence of MMT on the drug loading, and encapsulation efficiency properties of CS/MMT microspheres are shown in Fig. 6. Obviously, the drug incorporation process via solvent exchange is much more efficient than passive adsorption. The adsorption property of chitosan microspheres is mainly due to its large surface and swelling property [47]. However, the dried chitosan microspheres are hardly swell in ethanol, so the loading efficiency of passive adsorption is very low. The advantage of adsorbing drug by solvent exchange process during microspheres dehydration arises that it overcomes the disadvantage of the poor solubility of TA in water and the poor adsorption capacity of dried chitosan microspheres in ethanol. It greatly improves the efficiency of drug loading, and successfully incorporates hydrophobic drugs into hydrophilic matrix.

As seen in Fig. 6, the EE of the TA in pure CS microspheres is 36.09%, and with the weight ratio of the CS to MMT increased from 10:1 to 10:2, the EE of the microspheres increased from 44.50 to 48.18%, but then decreased with the further increase of MMT. The tendency of DL is in line

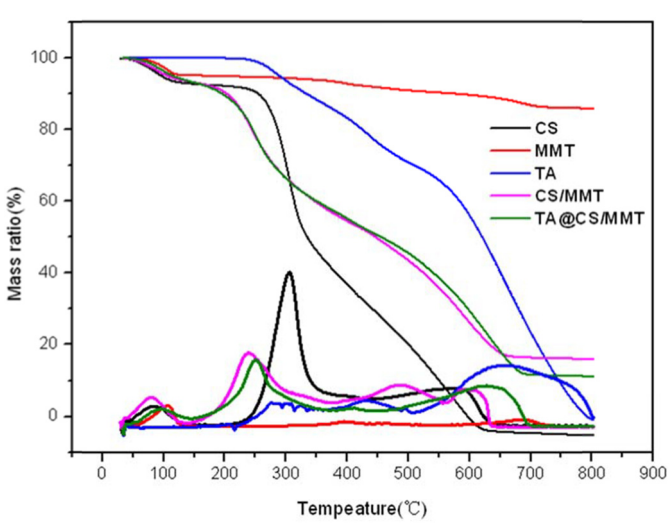


Fig. 5. TGA curves of TA, CS, MMT, CS/MMT, and TA@CS/MMT.

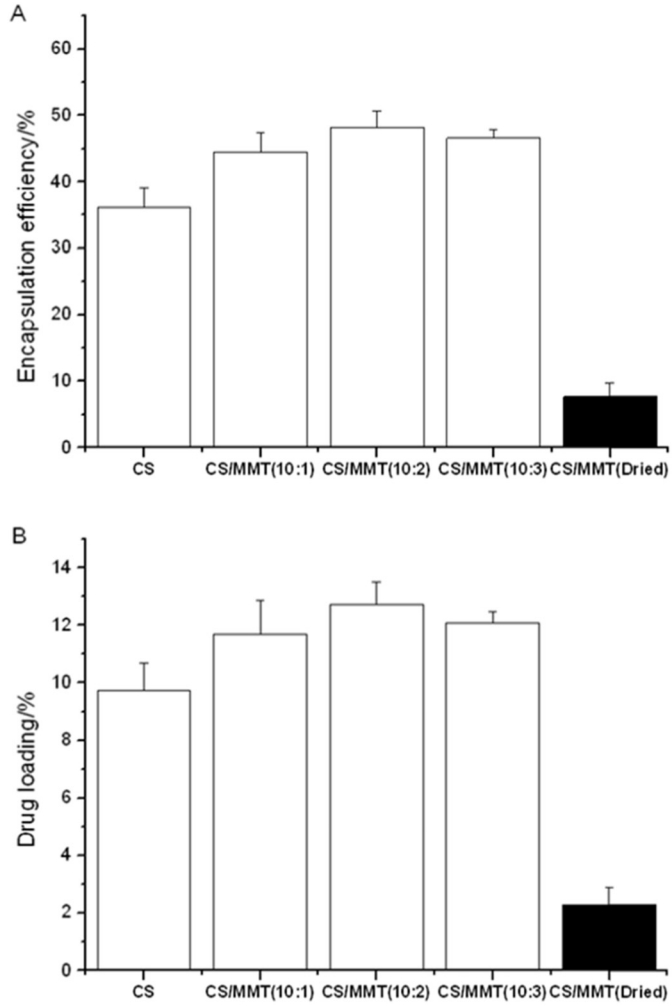


Fig. 6. Effect of MMT content on encapsulation efficiency (A) and drug loading (B) of CS/MMT microspheres.

with the EE. The drug loading property of CS/MMT microspheres in this study is mainly controlled by adsorption and diffusion in the process of solvent exchange [15]. The increase of EE may be due to the enhanced interaction between TA and CS/MMT, such as the improved network of the microspheres, which is favorable to the encapsulation of drug molecules. In the CS/MMT microspheres, there are two crosslinks responsible for the density of the network: *i)* the physical crosslinking between CS and MMT layers by electrostatic interaction, and *ii)* the chemical crosslinking between CS and the glutaraldehyde [48]. The TA molecule can enter the network of microspheres with the solvent exchange process, and shuttles between the microsphere and the solution. The physical crosslinking and chemical crosslinking interactions in combine construct the three-dimensional network of the microspheres which can restrict the mobility and diffusion of TA from the CS/MMT. In addition to the three-dimensional network structure, the TA molecule can also enter the interlayer space of MMT, which will help to improve the EE. However, the higher amount of MMT will fill in the matrix and leads to the less free space of the microspheres which eventually limits the penetration of TA or ethanol molecules into the microspheres. Thus, higher MMT content results in the decline of EE of CS/MMT microspheres.

3.3. In vitro release studies

In vitro release of TA from TA@CT/MMT microspheres was investigated by the dialysis bag diffusion technique in buffer solutions (pH 7.4

PBS and containing 0.5% sodium dodecyl sulfate) at 37 °C. The relationship between percentage of cumulative release and incubation time is shown in Fig. 7. Obviously, the MMT proportion is an important factor that influences the release rate, as the ratio of MMT goes from 10:0 to 10:3, the release rate gradually decreases. The in vitro release of TA@CS and TA@CS/MMT microspheres represents similar release profiles, which exhibits a small burst release in the first 4 h, followed by gradually slower down in the subsequent time but with different rates. The initial burst effect may be attributed to the fast release of the drug adsorbed on the surface of the microspheres. The mechanism of the release rate affected by the MMT proportion was similar to the EE, the much greater quantity of MMT in the matrix which gives rise to a compact network of biopolymer chains, and the greater interaction between the physical crosslinking and chemical crosslinking resulted in the restrained mobility and diffusion of TA from the microspheres. Furthermore, due to the lamellar structure of MMT, the TA molecule can enter the interlayer structure of MMT and increase the adsorption capacity of matrix, so increasing MMT ratio will help to reduce the release rate. But, with further increase in the MMT proportion from 10:2 to 10:3, no obvious difference was observed, the tendency was similar with the results of Fig. 6A showing EE of the CS/MMT microspheres with the same content of MMT. Although the improvement of MMT proportion can promote the adsorption capacity by increasing the probability of TA molecules enter the interlayer structure of MMT, but the higher amount of MMT in the matrix will limit the penetration of TA into the microspheres, and which eventually affect the enhancement of adsorption capacity. In summary, the incorporation of MMT into CS matrix can significantly improve the drug encapsulation and sustained-release properties of CS microspheres, and the bioavailability improvement of TA is superior to previous work [49].

3.4. In vitro drug release kinetics

For the mathematical evaluations, the release kinetics of TA from TA@CS/MMT microspheres were determined by fitting the in vitro TA release data to standard release equations (zero order, first order, Higuchi, and Korsmeyer–Peppas model) [40]. The analysis of determination coefficients (R^2) of linear relationship between TA release and time is established for the evaluation of the kinetic model. The kinetic parameters obtained are shown in Table 1. The release data of all tested samples show proper fitting to Higuchi model evidenced by the higher R^2 values (0.9636–0.9668) compared to those obtained from the other calculated mathematical models. To figure out the mechanism that controls the drug release, first 60% drug release data are fitted in Korsmeyer-Peppas model, the ln Q is plotted as a function of ln t only with the data until 10 h in this study, the n value indicated different release mechanisms [50]. For the case of spherical matrices, the value of n below 0.43 indicates that the drug release from polymer follows Fickian diffusion, 0.43–0.85 is an indicator of non-Fickian transport and above 0.85 is considered as case II transport [41]. As is present in Table 1, the n value is listed between (0.4696–0.5352), indicating that the TA release mechanism is non-Fickian diffusion. These results indicate that the release process contains both diffusion controlled drug release and corrosion controlled drug release [51]. And the results also imply no variation in the release mechanism upon MMT addition.

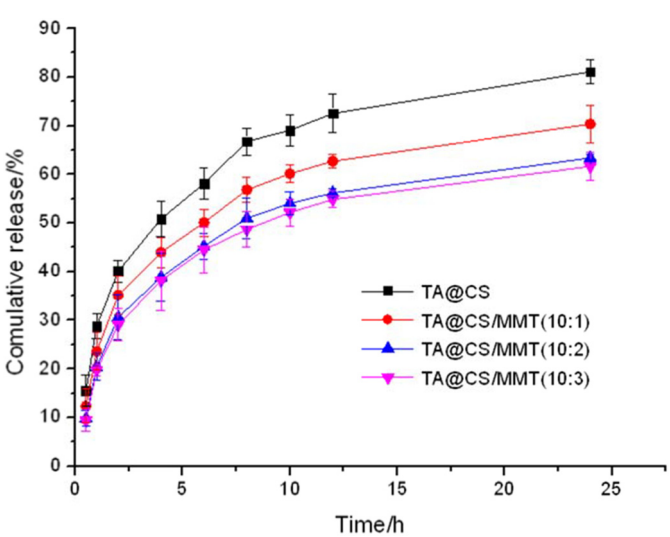


Fig. 7. Effect of MMT content on the release behavior of tanshinone IIA from microspheres.

3.5. Cellular viability

The cytotoxicity of the microspheres with or without TA were evaluated in the Caco-2 cell line. As shown in Fig. 8, the cell viability is assessed at different concentrations of CS/MMT microspheres or TA@CS/MMT microspheres after 24 h of incubation. In the case of the microspheres without TA (Fig. 8A), a significant decrease ($p < 0.05$) in the viability is observed at the highest concentrations (100 μg/ml), but not in the groups with lower concentrations of treatment, indicating excellent biocompatibility and negligible cytotoxicity of the drug delivery. In parallel, the results also indicate no effect of MMT on cell viability, suggesting that the combine use of MMT and CS to form the composite materials are biocompatible at the studied MMT proportion, similar as the results from a previous study [52].

The TA@CS/MMT (10:2), which has the highest EE, is used for the cytotoxicity evaluation of the drug-loaded microspheres. The concentration of samples is consistent with CS/MMT microspheres, and the pure drug (TA) at 5 μg/ml is treated as the model control group, equivalent to the amount of TA in TA@CS/MMT (10:2) microspheres. As seen in Fig. 8B, the tendency of TA@CS/MMT (10:2) is quite similar as the CS/MMT microspheres without TA. These results indicate that the drug-loaded microspheres also have ideal biocompatibility at the studied concentrations.

4. Conclusions

In this study, the TA loaded chitosan-montmorillonite microspheres with the sustained release behavior were successfully prepared by Emulsion crosslinking and solvent exchange process. We demonstrated that the properties of the composite microspheres significantly depended on the amount of the MMT. On the one hand, the introduced MMT can affect the drug loading and release performance by impacting the network of chitosan matrix. On the other hand, the MMT can also promote the adsorption capacity by its interlayer structure. And the MMT has no effect on cell viability of the composite microspheres. The drug release process was better fitted by Higuchi model, and the release process contains both diffusion and corrosion. The well properties of the composite microspheres reveal a promising prospect for its applications as drug delivery system. The method of drug adsorption was carried out by using the solvent exchange process of chitosan matrix dehydration is straightforward and effective. This study supplies another excellent route for the encapsulation of hydrophobic drugs.

Table 1
Kinetic parameter for tanshinone IIA release from TA@CS/MMT microspheres.

Microspheres code	Zero order	First order	Higuchi	Korsmeyer-Peppas	
	R^2	R^2	R^2	R^2	n
TA@CS	0.8836	0.9585	0.9668	0.9609	0.4696
TA@CS/MMT(10:1)	0.8778	0.9423	0.9636	0.9536	0.4952
TA@CS/MMT(10:2)	0.8771	0.9327	0.9636	0.9477	0.5307
TA@CS/MMT(10:3)	0.8770	0.9313	0.9642	0.9469	0.5352

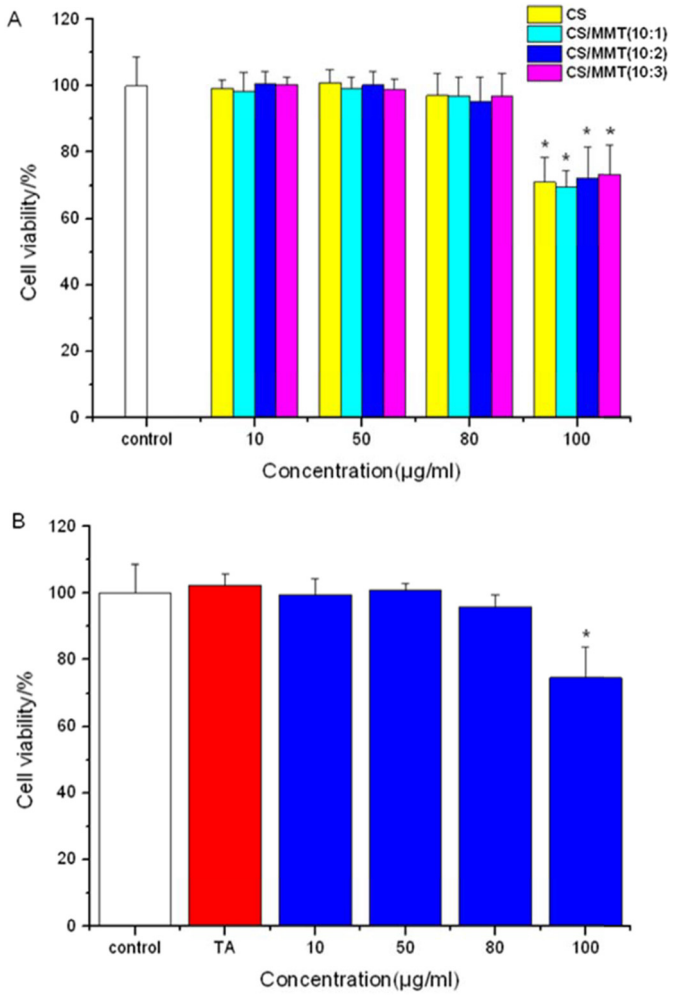


Fig. 8. Caco-2 cell viability (%) after contact with different concentration of CS/MMT microspheres (A) and TA@CS/MMT(10:2) (B). Data shows the mean ± SD (n = 5). *P < 0.05 compared with control.

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References

[1] S. Gao, Z.P. Liu, H. Li, P.J. Little, P.Q. Liu, S.W. Xu, Cardiovascular actions and therapeutic potential of tanshinone IIA, Atherosclerosis 220 (2012) 3–10.
[2] J.Y. Han, J.Y. Fan, Y. Horie, S. Miura, D.H. Cui, H. Ishii, T. Hibi, H. Tsuneki, I. Kimura, Ameliorating effects of compounds derived from *Salvia miltiorrhiza* root extract on microcirculatory disturbance and target organ injury by ischemia and reperfusion, Pharmacol. Ther. 117 (2008) 280–295.
[3] T. Song, Y. Yao, T. Wang, H. Huang, H. Xia, Tanshinone IIA ameliorates apoptosis of myocardiocytes by up-regulation of miR-133 and suppression of Caspase-9, Eur. J. Pharmacol. 815 (2017) 343–350.
[4] X.H. Wang, S.L. Morris-Natschke, K.H. Lee, New developments in the chemistry and biology of the bioactive constituents of Tanshen, Med. Res. Rev. 27 (2007) 133–148.
[5] J.G. Wang, S.C. Bondy, L. Zhou, F.Z. Yang, Z.G. Ding, Y. Hu, Y. Tian, P.Y. Wen, H. Luo, F. Wang, W.W. Li, J. Zhou, Protective effect of tanshinone IIA against infarct size and increased HMGB1, NFkappaB, GFAP and apoptosis consequent to transient middle cerebral artery occlusion, Neurochem. Res. 39 (2014) 295–304.
[6] X. Liu, M. Ye, C.Y. An, L.Q. Pan, L.T. Ji, The effect of cationic albumin-conjugated PEGylated tanshinone IIA nanoparticles on neuronal signal pathways and neuroprotection in cerebral ischemia, Biomaterials 34 (2013) 6893–6905.
[7] S.C. Chiu, S.Y. Huang, S.P. Chen, C.C. Su, T.L. Chiu, C.Y. Pang, Tanshinone IIA inhibits human prostate cancer cells growth by induction of endoplasmic reticulum stress in vitro and in vivo, Prostate Cancer Prostatic Dis. 16 (2013) 315–322.
[8] C. Lin, L. Wang, H. Wang, L. Yang, H. Guo, X. Wang, Tanshinone IIA inhibits breast cancer stem cells growth in vitro and in vivo through attenuation of IL-6/STAT3/NF-κB signaling pathways, J. Cell. Biochem. 114 (2013) 2061–2070.
[9] H.X. Yan, J. Li, Z.H. Li, W.L. Zhang, J.P. Liu, Tanshinone IIA – loaded pellets developed for angina chronotherapy: deconvolution-based formulation design and optimization, pharmacokinetic and pharmacodynamic evaluation, Eur. J. Pharm. Sci. 76 (2015) 156–164.
[10] W.L. Zhang, H.L. He, J.P. Liu, J. Wang, S.Y. Zhang, S.S. Zhang, Z.M. Wu, Pharmacokinetics and atherosclerotic lesions targeting effects of tanshinone IIA discoidal and spherical biomimetic high density lipoproteins, Biomaterials 34 (2013) 306–319.
[11] Z. Meng, L. Meng, K. Wang, J. Li, X. Cao, J. Wu, Y. Hu, Enhanced hepatic targeting, biodistribution and antifibrotic efficacy of tanshinone IIA loaded globin nanoparticles, Eur. J. Pharm. Sci. 73 (2015) 35–43.
[12] J.M. Zhang, Y.B. Li, X.F. Fang, D.M. Zhou, Y.T. Wang, M.W. Chen, TPGS-g-PLGA/Pluronic F68 mixed micelles for tanshinone IIA delivery in cancer therapy, Int. J. Pharm. 476 (2014) 185–198.
[13] J. Li, P. Liu, J.P. Liu, W.L. Zhang, J.K. Yang, Y.Q. Fan, Novel tanshinone II A ternary solid dispersion pellets prepared by a single-step technique: in vitro and in vivo evaluation, Eur. J. Pharm. Biopharm. 80 (2012) 426–432.
[14] H. Ma, Q. Fan, J. Yu, J. Xin, C. Zhang, Novel microemulsion of tanshinone IIA, isolated from *Salvia miltiorrhiza* Bunge, exerts anticancer activity through inducing apoptosis in hepatoma cells, Am. J. Chin. Med. 41 (2013) 197–210.
[15] J.H. Xu, H. Zhao, W.J. Lan, G.S. Luo, A novel microfluidic approach for monodispersed chitosan microspheres with controllable structures, Adv. Healthc. Mater. 1 (2012) 106–111.
[16] S. Qian, Q.Z. Zhang, Y.F. Wang, B. Lee, G.V. Betageri, M.S. Chow, M. Huang, Z. Zuo, Bioavailability enhancement of glucosamine hydrochloride by chitosan, Int. J. Pharm. 455 (2013) 365–373.
[17] A. Bernkop-Schnürch, S. Dünnhaupt, Chitosan-based drug delivery systems, Eur. J. Pharm. Biopharm. 81 (2012) 463–469.
[18] X. Sun, Z.H. Tang, M. Pan, Z.C. Wang, H.Q. Yang, H.Q. Liu, Chitosan/kaolin composite porous microspheres with high hemostatic efficacy, Carbohydr. Polym. 177 (2017) 135–143.
[19] P. Chanphai, H.A. Tajmir-Riahi, Encapsulation of testosterone by chitosan nanoparticles, Int. J. Biol. Macromol. 98 (2017) 535–541.
[20] B. Huang, M.X. Liu, C.R. Zhou, Chitosan composite hydrogels reinforced with natural clay nanotubes, Carbohydr. Polym. 175 (2017) 689–698.
[21] J.A. Vilela, F.de A. Perrechil, C.S. Picone, A.C. Sato, R.L. da Cunha, Preparation, characterization and in vitro digestibility of gellan and chitosan–gellan microgels, Carbohydr. Polym. 117 (2015) 54–62.
[22] Q. Liu, M.D. Xu, Y.D. Wang, R.K. Feng, Z. Yang, S.F. Zuo, C.Z. Qi, M.F. Zeng, Co-immobilization of Pd and Zn nanoparticles in chitosan/silica membranes for efficient, recyclable catalysts used in Ullmann reaction, Int. J. Biol. Macromol. 105 (2017) 575–583.
[23] Natália G. Graebin, Diandra de Andrades, Marina C. Bonin, Rafael C. Rodrigues, Marco A.Z. Ayub, Dextransucrase immobilized on activated-chitosan particles as a novel biocatalyst, J. Mol. Catal. B Enzym. 133 (2016) S143–S149.
[24] J.F. Zhang, Q. Tang, X.Y. Xu, N. Li, Development and evaluation of a novel phytosome-loaded chitosan microsphere system for curcumin delivery, Int. J. Pharm. 448 (2013) 168–174.
[25] S. Jayrajsinh, G. Shankar, M. Pharm, Y.K. Agrawal, L. Bakre, Montmorillonite nanoclay as a multifaceted drug-delivery carrier: a review, J. Drug Delivery Sci. Technol. 39 (2017) 200–209.
[26] C. Stathokostopoulou, P.A. Tarantili, Preparation, characterization and drug release studies from poly (d, l-lactic acid)/organoclay nanocomposite films, J. Macromol. Sci. A 51 (2014) 117–124.
[27] L. Bounabi, N.B. Mokhnachi, N. Haddadine, F. Ouazib, R. Barille, Development of poly (2-hydroxyethyl methacrylate)/clay composites as drug delivery systems of paracetamol, J. Drug Delivery Sci. Technol. 33 (2016) 58–65.
[28] N. Salahuddin, R. Abdeen, Drug release behavior and antitumor efficiency of 5- ASA loaded chitosan-layered silicate nanocomposites, J. Inorg. Organomet. Polym. Mater. 23 (2013) 1078–1088.
[29] B.D. Kevadiya, S.S. Chettiar, S. Rajkumar, H.C. Bajaj, K.A. Gosai, H. Brahmbhatt, Evaluation of clay/poly(L-lactide) microcomposites as anticancer drug, 6-mercaptopurine reservoir through in vitro cytotoxicity, oxidative stress markers and in vivo pharmacokinetics, Colloids Surf. B: Biointerfaces 112 (2013) 400–407.
[30] H. Kaygusuz, M. Uysal, V. Adımcılar, F.B. Erim, Natural alginate biopolymer montmorillonite clay composites for vitamin B2 delivery, J. Bioact. Compat. Polym. 30 (2015) 48–56.
[31] Y. Huang, Q. Tao, D. Hou, S. Hu, S. Tian, Y. Chen, R. Gui, L. Yang, Y. Wang, A novel ion-exchange carrier based upon liposome-encapsulated montmorillonite for ophthalmic delivery of betaxolol hydrochloride, Int. J. Nanomedicine 12 (2017) 1731–1745.
[32] V. Iannuccelli, E. Maretti, M. Montorsi, C. Rustichelli, F. Sacchetti, E. Leo, Gastroretentive montmorillonite-tetracycline nanoclay for the treatment of *Helicobacter pylori* infection, Int. J. Pharm. 493 (2015) 295–304.
[33] I. Salcedo, C. Aguzz, G. Sandri, M.C. Bonferoni, M. Mori, P. Cerezo, R. Sánchez, C. Viseras, C. Caramella, In vitro biocompatibility and mucoadhesion of montmorillonite chitosan nanocomposite: a new drug delivery, Appl. Clay Sci. 55 (2012) 131–137.
[34] S. Bensalema, B. Hamdi, S. Del Confetto, M. Iguer-Ouada, A. Chamayoud, H. Balardf, R. Calvet, Characterization of chitosan/montmorillonite bionanocomposites by inverse gas chromatography, Colloids Surf. A Physicochem. Eng. Asp. 516 (2017) 336–344.
[35] P. Lertsutthiwongn, K. Noomun, S. Khunthon, S. Limpanart, Influence of chitosan characteristics on the properties of biopolymeric chitosan–montmorillonite, Prog. Nat. Sci.: Mater. Int. 22 (2012) 502–508.
[36] S.B. Patil, K.K. Sawant, Chitosan microspheres as a delivery system for nasal insufflation, Colloids Surf. B. Biointerfaces 84 (2011) 384–389.
[37] K.G. Desai1, H.J. Park, Preparation and characterization of drug-loaded chitosan–tripolyphosphate microspheres by spray drying, Drug Dev. Res. 64 (2005) 114–128.

[38] H. Ennajih, R. Bouhfid, E.M. Essassi, M. Bousmina, A.E. Kadib, Chitosan–montmorillonite bio-based aerogel hybrid microspheres, Microporous Mesoporous Mater. 152 (2012) 208–213.
[39] J. Wu, S.J. Ding, J. Chen, S.Q. Zhou, H.Y. Ding, Preparation and drug release properties of chitosan/organomodified palygorskite microspheres, Int. J. Biol. Macromol. 68 (2014) 107–112.
[40] A.A. Kassem, S.H. Abd El-Alim, M. Basha, A. Salama, Phospholipid complex enriched micelles: a novel drug delivery approach for promoting the antidiabetic effect of repaglinide, Eur. J. Pharm. Sci. 99 (2017) 75–84.
[41] J. Siepmann, N.A. Peppas, Modeling of drug release from delivery systems based on hydroxypropyl methylcellulose (HPMC), Adv. Drug Deliv. Rev. 48 (2001) 139–157.
[42] B.R. dos Santos, F.B. Bacalhau, S. Pereira Tdos, C.F. Souza, R. Faez, Chitosan-montmorillonite microspheres: a sustainable fertilizer delivery system, Carbohydr. Polym. 127 (2015) 340–346.
[43] S. Frindy, A. Primo, A. el K. Qaiss, R. Bouhfid, M. Lahcini, H. Garcia, M. Bousmina, A. El Kadib, Insightful understanding of the role of clay topology on the stability of biomimetic hybrid chitosan-clay thin films and CO_2-dried porous aerogel microspheres, Carbohydr. Polym. 146 (2016) 353–361.
[44] M.M. Beppu, R.S. Vieira, C.G. Aimoli, C.C. Santana, Crosslinking of chitosan membranes using glutaraldehyde: effect on ion permeability and water absorption, J. Membr. Sci. 301 (2007) 126–130.
[45] K.H. Liu, T.Y. Liu, S.Y. Chen, D.M. Liu, Drug release behavior of chitosan–montmorillonite nanocomposite hydrogels following electrostimulation, Acta Biomater. 4 (2008) 1038–1045.
[46] D. Depan, A.P. Kumar, R.P. Singh, Cell proliferation and controlled drug release studies of nanohybrids based on chitosan-g-lactic acid and montmorillonite, Acta Biomater. 5 (2009) 93–100.
[47] G.Z. Kyzas, M. Kostoglou, Swelling–adsorption interactions during mercury and nickel ions removal by chitosan derivatives, Sep. Purif. Technol. 149 (2015) 92–102.
[48] S.B. Hua, H.X. Yang, W.B. Wang, A.Q. Wang, Controlled release of ofloxacin from chitosan–montmorillonite hydrogel, Appl. Clay Sci. 50 (2010) 112–117.
[49] Q.Y. Liu, Z.H. Zhang, X. Jin, Y.R. Jiang, X.B. Jia, Enhanced dissolution and oral bioavailability of tanshinone IIA base by solid dispersion system with low-molecular-weight chitosan, J. Pharm. Pharmacol. 65 (2013) 839–846.
[50] P.L. Ritger, N.A. Peppas, A simple equation for description of solute release I. Fickian and non-fickian release from non-swellable devices in the form of slabs, spheres, cylinders or discs, J. Control. Release 5 (1987) 23–36.
[51] L.L. Lao, N.A. Peppas, F.Y. Boey, S.S. Venkatraman, Modeling of drug release from bulk-degrading polymers, Int. J. Pharm. 418 (2011) 28–41.
[52] I. Salcedo, G. Sandri, C. Aguzzi, C. Bonferoni, P. Cerezo, R. Sánchez-Espejo, C. Viseras, Intestinal permeability of oxytetracycline from chitosan-montmorillonite nanocomposites, Colloids Surf. B: Biointerfaces 117 (2014) 441–448.